A PRODUCT MANAGER'S FIELD GUIDE

Shelfmate

Your skincare routine, built in a minute.

Answer three questions. Get a routine built only from the shelf.
See the whole shop through the merchant's eyes.

What it is, how it works, why it is built this way,
and how to talk about it in an interview.

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How to read this: every chapter opens with the plain-English idea, then shows how it really works, then tells you what a PM should take from it. Coloured boxes mark analogies, PM lenses and warnings. You can read it front to back in about an hour, or jump to chapter 11 (cheat sheet) right before an interview.

CHAPTER 01 · START HERE

The 60-second version

Shelfmate is a small online skincare shop with two faces. Shoppers answer three questions and get a routine built only from the shelf. The merchant gets a dashboard that shows where shoppers drop off and what a discount really does to margin.

2 views: shopper and merchant	**12** products in the invented catalog	**0** made-up products that can reach a cart

What it is, and who it is for

Shelfmate is a concept project for a made-up Indian direct-to-consumer skincare brand. Nothing is sold, and no payment, name or address is collected. Orders are demo orders. It was built to show end-to-end product thinking: the **shopper journey**, the **merchant's view of the business**, and **AI used safely** in the middle.

Who	Their job to be done	What Shelfmate gives them
Shopper	"I don't know what to buy, and I'm scared of picking the wrong thing"	A routine of 3 or 4 steps, one product per step, within budget, gentle when needed, each with a reason
Merchant	"Where am I losing people, and is my discount worth it?"	Funnel, best sellers, recent orders, a plain-language insight and a discount what-if

WHY THIS PROJECT EXISTS

Most AI shopping demos stop at a clever chat. Shelfmate shows the harder half: what happens **after** the AI answers. The answer is validated, priced on the server, tracked as a funnel, and turned into decisions for the merchant. That whole loop is the product.

The one-sentence pitch

"A guided skincare routine builder where the AI can only recommend what is really on the shelf, every answer is checked before the shopper sees it, and the merchant gets the numbers to run the shop."

The problem, in plain words

ANALOGY

Walk into a skincare shop with fifty bottles. You freeze. A good shop assistant asks three questions, then hands you a cleanser, a serum, a moisturiser and a sunscreen, and tells you why. Shelfmate is that assistant, online. The merchant is the shop owner watching the door counter and asking "how many people walked in, how many left, and why?"

The shopper's headache

- **Too many choices.** Skincare shelves are overwhelming and the words (niacinamide, ceramide, bakuchiol) mean nothing to most people.
- **Fear of a wrong or harsh product.** Sensitive skin, pregnancy or a prescription make a bad pick feel risky.
- **Budget is real.** A routine that quietly costs double what you planned loses the sale.

The merchant's headache

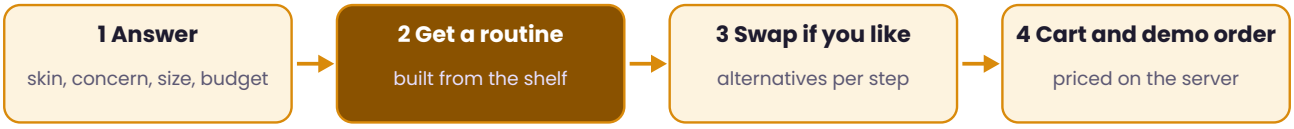
- **Invisible drop-off.** Visitors vanish somewhere between landing and paying, and nobody knows where.
- **Discounts are guesses.** A bundle discount lifts sales but eats margin. By how much?
- **Shoppers say things the shop never hears.** The free-text note ("fragrance-free please") holds product ideas, but nobody reads it.

PM LENS

Two-sided thinking is the point. A recommender that makes shoppers happy but gives the merchant no way to learn from the data is a feature. One that closes the loop is a product. Notice how each shopper action in this app creates a data point the merchant later reads.

CHAPTER 03 · WHAT A VISITOR SEES

The shopper journey



Step 1: Answer three questions (plus a note)

Question	Choices
Skin type	Oily, Dry, Combination, Sensitive
Main concern	Breakouts, Dullness and dark spots, Dryness, Fine lines, Redness
Routine size	3 steps (Cleanse, Moisturize, Protect) or 4 steps (add a Treat step)
Budget	Up to ₹1,500, up to ₹2,500, or no limit
Optional note	Free text, up to 400 characters. Emails, phone numbers and links are stripped

Step 2: Get a routine

You get one product per step, each with a short reason that refers to your answers, a total price, and a list of **safety checks** that passed (every product from the catalog, one per step, within budget, and gentle products only when needed). A small label tells you whether the routine came from Claude, Gemini or the simple rules.

Step 3: Swap any step

Do not like the serum? Swap it for another product from the same step. The server checks the swap is valid (same step, a real product, and gentle if the routine must be gentle) and reprices the routine. You can also open **Browse the whole shelf** and add single products.

Step 4: Cart and demo order

Add the routine (as one bundle) or single products to the cart and place a demo order. The browser never sends a price. The server looks up each product, applies the routine discount if the bundle is valid, and stores the order.

TRY IT YOURSELF

Pick Sensitive skin, then in the note type "I am pregnant". Notice the routine changes to gentle-only products and a see-a-professional note appears. Then try a tight budget with 4 steps and read the honest message: the cheapest 4-step routine here is ₹1,696, above ₹1,500.

PM LENS: HONEST EDGE CASES

Many demos hide the case where the user's wish is impossible. Shelfmate says it plainly: with this catalog a 4-step routine cannot fit under ₹1,500, so it suggests 3 steps or a bigger budget. Telling the truth at the constraint builds more trust than quietly breaking the budget.

The merchant view

The merchant page (`/merchant`) is a small analytics dashboard. It turns shopper events into answers.

Block	What it shows	Why a merchant cares
Four headline numbers	Routines built, orders and revenue, conversion, average order value	Health of the shop at a glance
What stands out	A plain-language line: the biggest drop-off, and routine vs no routine (labelled as a comparison, not proof)	Turns charts into a next question
Funnel	Sessions, started quiz, finished quiz, viewed routine, added to cart, purchased	Shows exactly where people leave
Best sellers and recent orders	Top products by orders; latest orders	What to stock and promote
Most swapped products	Which products shoppers replace most	A product or fit problem
What shoppers ask for	The free-text notes grouped into five themes, each with a suggested action	Free product research
Average order value by day	Last 8 days	Spot trends and spikes
Discount switch and what-if	Set the routine discount and see revenue and margin per session at different levels	Price with eyes open

The funnel, explained

A funnel counts how many people reach each step. For example, if 100 visitors arrive and 42 start the quiz, 31 finish, 31 see a routine, 16 add to cart and 7 buy, the biggest leak is between starting and finishing or between cart and purchase, depending on the ratios. The dashboard highlights the biggest drop so the merchant knows where to look first.

WORKED EXAMPLE (INVENTED NUMBERS)
Sessions 100 → Started 42 (42%) → Finished 31 (74% of started) → Added to cart 16 (52% of finished) → Purchased 7 (44% of carts).
Overall conversion is 7 / 100 = **7%**. A merchant who improves the cart-to-purchase step from 44% to 60% gains 2 extra orders for every 100 sessions, with no extra traffic.

What shoppers ask for: turning notes into ideas

Free-text notes are matched against five themes using simple patterns, and each theme carries a suggested action.

Theme	Example words	Suggested action
Acne marks and dark spots	acne marks, dark spot, pigmentation	Lead with a brightening pairing in the quiz results

Theme	Example words	Suggested action
Fragrance-free	fragrance, perfume, scent	Add a fragrance-free filter to the quiz
Seasonal change	summer, winter, monsoon, humid	Offer a seasonal swap for the moisturiser
Pregnancy or a skin condition	pregnant, allergy, prescription	Keep the gentle routine and the professional note
Budget	cheap, afford, under ₹500	Add a two-product starter option

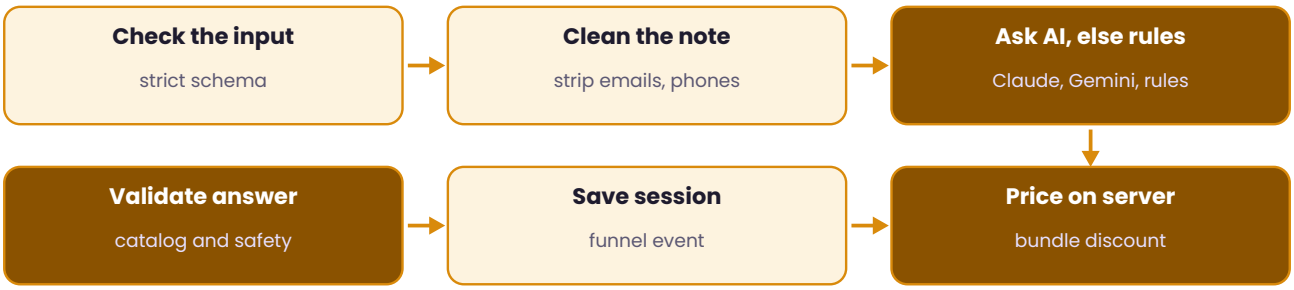
HONEST SAMPLE DATA

A new shop dashboard with zero data looks broken. So the first visit adds **80 clearly flagged demo sessions**, generated by a seeded random generator so they are the same every time. A checkbox hides them, and real and demo counts are always shown. Conversion from a handful of real visitors means very little, and the README says so.

CHAPTER 05 · THE MACHINERY, SIMPLY

Under the hood: how it works

Shelfmate is a Next.js app deployed on Vercel. The browser sends small requests to server routes. The server does anything that touches money, safety or data. Here is the journey of one routine request.



Piece	What it does	In plain words
Catalog (catalog.ts)	12 invented products with step, price, skin types, concerns and a gentle flag	The shelf
Rules routine (shop.ts)	Scores each product for the shopper, picks the best per step, swaps to cheaper ones until the budget fits	The reliable assistant
AI prompt and validator	Asks a model for JSON, then rejects anything that fails the checks	The strict supervisor
Routes: routine, track, order	Build routines, log events and swaps, place demo orders	The three counters in the shop
Session store (sessions.ts)	One JSON record per visitor session: answers, routine, swaps, cart, purchase	The notebook the merchant reads
Stats (stats.ts)	Funnel, conversion, average order, top products, what-if maths	The accountant
Limits and redaction	Hourly limits, daily AI cap, personal-data stripping	The security guard
Storage (store.ts)	Vercel Blob in production, in-memory locally	The filing cabinet

The server is the authority

A recurring design principle: the browser is never trusted for anything that matters. Prices are computed on the server. A swap is only accepted if the server agrees it is valid. A bundle is only a bundle if it has exactly one catalog product per step. This is what stops a curious visitor, or a buggy model, from creating a cheap or impossible order.

> **PM LENS**
> "Where does the source of truth live?" is a question PMs should ask of every feature. Here the answer for price, validity and safety is always the server. That is why the demo can be public without people editing totals in their browser.

How the routine is built

Path A: the rules routine (no AI)

For each step the rules look only at products in that step (and only gentle ones if the shopper needs gentle). Each product gets a score, and the highest score wins:

score = 3 if it suits your skin type + 3 if it targets your concern - price / 10,000

The tiny price term only breaks ties in favour of the cheaper product. If the total is over budget, the rules look for the single swap that saves the most money (within the same step) and repeat until the routine fits or nothing cheaper exists.

WORKED EXAMPLE (ILLUSTRATIVE)

Oily skin, concern: breakouts, step: Cleanse. **Neem Clarify Cleanser** suits oily (+3) and targets breakouts (+3), price ₹449: score about 5.96. **Gentle Gel Cleanser** suits oily (+3) but does not target breakouts, price ₹399: score about 2.96. Neem wins.

Now the shopper adds "I have eczema". The note matches the risk pattern, so only **gentle** products are allowed. Neem is not gentle, so Gentle Gel Cleanser is picked instead.

Path B: the AI routine

The model gets the shopper's answers and the whole catalog as data, and is told: pick exactly one product per step using only catalog ids, stay within budget, give each reason in 22 words or fewer, make no medical claims, and if the note mentions pregnancy, breastfeeding, allergy, prescription or a skin condition, choose only gentle products. The shopper's note is passed as *data, not instructions*.

The safety net: four checks

Check	What it protects against
Every product is from the catalog	A model inventing a product that does not exist
One product per step, in the right order	Two serums and no sunscreen
Within budget	Silent overspending
Gentle products only (when needed)	A harsh active in a pregnancy or sensitive-skin routine

If a model's answer fails any check, it is thrown away. The next provider gets a turn (Claude, then Gemini). If all fail or AI is off or capped, the rules routine is used and a short note tells the shopper why.

THE KEY IDEA

The model proposes, the code disposes. The model is creative, but a boring piece of code decides whether the output is allowed through. That single design choice is why a made-up or unsafe product can never reach a cart.

ANALOGY

Think of the AI as a keen new stylist and the validator as the shop manager who checks each outfit against the stock list and the customer's notes before it leaves the fitting room.

The risk pattern

The words pregnant, breastfeeding, nursing, allergy, eczema, rosacea, psoriasis, prescription, isotretinoin, accutane, medication and dermatitis switch the routine to gentle-only and add a see-a-professional note. It is a deliberately simple, visible rule. It will miss some phrasing, which is why the note recommends professional advice rather than claiming safety.

Pricing, discounts and margin

Bundle discount

A routine of 3 or 4 products qualifies for the **routine discount** (10% by default). The discount is rounded to whole rupees. A single product never gets it. The merchant can change the percentage on the dashboard and shoppers see the change immediately.

Margin, with an honest assumption

The code assumes the cost of goods is **45% of the list price**. That is an input, not a measured fact, and the README says so. Margin per order = (what the shopper paid - cost) / what the shopper paid.

Discount	Routine list price	Shopper pays	Cost at 45%	Margin
0%	₹2,046	₹2,046	₹921	55.0%
10%	₹2,046	₹1,841	₹921	50.0%
20%	₹2,046	₹1,637	₹921	43.8%

Example routine (illustrative): Gentle Gel Cleanser ₹399 + Niacinamide Serum ₹599 + Ceramide Barrier Cream ₹549 + SPF 50 Sheer Fluid ₹499 = ₹2,046.

The what-if

The dashboard recomputes revenue and margin per session as if every past bundle order had been priced at a different discount. It uses the same shoppers and the same baskets. This is a **mechanical what-if, not a forecast of behaviour**: it cannot tell you whether a deeper discount would bring more buyers, only what each order would have earned.

PM LENS: A DISCOUNT IS A TRADE

A bigger discount is only good if it brings enough extra orders to pay for the margin you give up. Using the table above, going from 10% to 20% cuts the margin on each order from 50.0% to 43.8% and the shopper pays ₹204 less. Profit per order falls from about ₹920 to ₹716, so you need roughly 29% more orders just to stand still on total profit. The what-if shows the first half of that trade, and an experiment (an A/B test) would be needed for the second half.

WATCH OUT

Do not read the what-if as "20% earns more". It holds the number of orders fixed. The honest label on the page is part of the product.

Safety, privacy and abuse protection

Layer	What it does	Why
Schema validation (zod)	Every request must match a strict shape: allowed skin types, sizes, budgets, id formats, lengths	Rejects malformed or hostile input
Free-text cap and redaction	Note capped at 400 characters; emails, phone numbers, links stripped	Keeps personal data away from models
Per-visitor hourly limits	Routines, orders, tracking, discount changes and demo-data setup each have a cap	Stops one visitor from draining the demo
Daily AI cap	Shared counter of model calls per day (default 400)	Hard ceiling on spend
Server-side pricing and validation	Totals and bundle validity computed on the server	No price tampering
Fallbacks	Rules routine always works	Limits never create a dead end
No payment, name or address	Orders are demos with only an id and product lines	Nothing sensitive to leak

HONEST LIMITS

The merchant view is open to anyone with the link, because this is a public demo. Anyone can change the discount, and the discount is the only setting it exposes. In a real shop this page would sit behind a login. Limits are tracked per server instance, which stops casual abuse but is not bank-grade.

Product decisions and trade-offs

Decision	Why	What we gave up
Three short questions, not a long quiz	Completion matters more than detail; every extra question loses people	Finer personalisation
Recommend only from the shelf	A recommendation you cannot buy is useless	Open-ended advice
Validate AI output in code	Trust is the product in a skincare context	Some good AI answers get rejected
Rules fallback	Always a working routine	Less tailored on that path
Server-side pricing	Prevents tampering, single source of truth	Slightly more round trips
Routine discount as the one merchant lever	Simple, visible and testable	No coupons, tiers or targeting
Flagged sample data	A dashboard that is never empty and never misleading	Demo numbers prove nothing
Demo order only	No payment risk, quick to try	Cannot test real checkout friction
Mechanical what-if instead of a forecast	Honest about what the data can say	No behaviour prediction

What a real product would add next

- A/B test the discount (0%, 10%, 15%) with real traffic and measure conversion and profit together.
- Real cost data instead of the 45% assumption.
- Login for the merchant view, and role-based access.
- Returning shopper history and replenishment reminders.
- A dermatologist-reviewed rule set for the gentle flag and the risk words.
- Payments, inventory and delivery, which this concept deliberately leaves out.

Metrics and testing

Layer	Metric	Question it answers
Funnel	Quiz start rate; quiz completion rate	Is the first screen inviting? Is the quiz short enough?
Value	Routine view rate; swap rate per product	Do shoppers accept the recommendation, or change it?
Commerce	Add-to-cart rate; conversion; average order value; bundle share	Is the routine selling the basket?
Pricing	Margin per session at each discount	Is the discount paying for itself?
Quality	AI pass rate vs rules fallback rate; check failures	How often does the AI earn trust?
Insight	Share of notes per theme	What should we build or stock next?

How it is tested

The automated tests (7 groups) check, among other things: the rules routine gives exactly one catalog product per step inside budget and passes all checks; a tight budget swaps to cheaper products; sensitive skin and risk notes force gentle products; **every combination** of skin, concern, size and budget yields a passing routine (the one impossible case, 4 steps under ₹1,500, must say so honestly); AI answers with non-catalog ids, missing steps, unsafe or over-budget picks are rejected; bundle pricing and validity; the dashboard maths; and basic catalog sanity.

> **PM LENS**
> Testing every combination of inputs is a product move, not only an engineering one: it proves there is no combination of answers that leaves a shopper with a broken screen.

Interview cheat sheet

The 30-second version

"Shelfmate is a skincare routine builder for an invented D2C brand. Shoppers answer three questions and get a routine built only from the shelf. The AI can propose, but code validates every answer against the catalog, the budget and a gentle-product rule, and falls back to simple rules if it fails. On the other side, a merchant dashboard shows the funnel, best sellers, what shoppers ask for, and what a different discount would do to margin."

The 2-minute version, in four beats

1. **Problem.** Shoppers are overwhelmed and afraid of a wrong product; merchants cannot see where people drop off or whether a discount pays.
2. **Solution.** A short quiz that produces a safe, priced routine, plus a merchant view that turns each shopper event into a number.
3. **Design choices.** Recommend only from the shelf; model proposes and code validates; server owns prices; honest labels on sample data and the what-if.
4. **Limits and next steps.** Margin uses an assumed 45% cost; the dashboard is open; the what-if is not a forecast. Next: an A/B test of discount levels, real cost data, a login.

Likely questions and good answers

Why not let the AI recommend freely?

A shopper cannot buy a product that is not on the shelf, and in skincare a wrong pick can be harmful. Restricting to the catalog and validating every answer makes the recommendation both buyable and safe.

How do you stop the AI recommending something unsafe?

Four checks run in code on every answer: catalog membership, one product per step, budget and, when needed, gentle-only. A failing answer is discarded and the next provider or the rules take over. The risk words in the note trigger the gentle rule and a see-a-professional message.

What does the merchant learn?

Where shoppers drop off, which products get swapped, what the notes ask for, what sells, and what a different discount would do to revenue and margin per session.

Is the discount what-if a forecast?

No. It re-prices the same baskets at a different percentage. It shows what each order would earn, not whether more people would buy. To know that you need an A/B test.

What assumptions are shaky?

The 45% cost of goods is an input, not a measurement; demo sessions are generated; and with few real visitors, conversion is noise. The README and the dashboard label all three.

How would you decide the discount?

Run an experiment with a few levels, measure conversion and profit per session together, and pick the level where profit per session is highest, not where conversion is highest.

What would you build next?

A discount A/B test, merchant login, real cost data, replenishment reminders for returning shoppers, and a clinician-reviewed safety rule set.

What did you cut on purpose?

Payments, accounts, inventory and delivery. They matter for a real shop but add nothing to the idea being shown, which is a safe AI recommendation loop and a merchant who can learn from it.

CHAPTER 12 · REFERENCE

Glossary and code map

Glossary

Term	Meaning
D2C	Direct to consumer: a brand selling straight to shoppers online
Routine / bundle	One product per step (3 or 4 steps), sold together
Step	Cleanse, Treat, Moisturize or Protect
Gentle	A product flag for sensitive skin and risk situations
Funnel	Counts of visitors reaching each stage, from visit to purchase
Conversion	Purchases divided by sessions
AOV	Average order value: revenue divided by orders
COGS / cost share	Cost of goods sold; here assumed 45% of list price
Margin	(Price paid - cost) / price paid
What-if	Re-pricing past orders at another discount, with behaviour held fixed
Validator	Code that accepts or rejects a model answer
Fallback	The simple rules used when AI is off, capped or rejected
Session	One visitor's record: answers, routine, swaps, cart and purchase

Where things live in the code

Repository: `apps/shelfmate` in the portfolio repo.

File	What is inside
lib/catalog.ts	The 12 products and the 45% cost share
lib/shop.ts	Options, risk words, rules routine, checks, AI validator, bundle discount, margin, AI prompt
lib/stats.ts	Dashboard maths, ask themes, what-if, seeded random generator
lib/sessions.ts	Session and config storage, loading, demo-session seeding
lib/ai.ts, limit.ts, redact.ts, store.ts	Provider chain, limits and daily cap, personal-data stripping, storage
app/api/routine, track, order, merchant	The server routes
components/Store.tsx, CartView.tsx, Merchant.tsx	Shopper screen, cart, merchant dashboard
tests/core.test.ts	The automated tests